6CS007 Project and Professionalism

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Abstract

HomeFix is a full-stack web application designed to address the significant gap in the home services market in Nepal. Existing platforms such as Sajilo Sewa and Smart Home Sewa provide limited flexibility, poor search functionality, and a narrow range of professional categories, leaving many users and skilled workers disconnected from one another. HomeFix aims to resolve these shortcomings by offering a dynamic, responsive platform where service consumers can search for, compare, and book professionals across an expanded range of categories which including traditional services such as plumbing and electrical work, as well as underserved categories like agricultural help, religious services, and IT support.

This report documents the design, development, and evaluation of the HomeFix system, covering the system architecture, database schema, frontend and backend implementation, testing strategy, and critical reflection on social, ethical, legal, and security considerations. The system has been built using a modern MERN stack (MongoDB, Express.js, React.js, and Node.js) and deployed via Vercel. Agile (Scrum) methodology was employed throughout the development lifecycle to ensure iterative improvement and stakeholder responsiveness. The report also evaluates the advanced artefact against the original project proposal and discusses deviations, technical decisions, and future enhancements.

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1. Introduction

1.1 Background and Motivation

The rapid pace of urbanisation in Nepal, particularly within the Kathmandu Valley, has created a growing demand for reliable, on-demand household services. Professionals in sectors such as plumbing, electrical repair, cleaning, IT support, and even agricultural assistance remain difficult to locate without relying on informal word-of-mouth networks or physical visits. Whilst this traditional approach has persisted for decades, it is increasingly insufficient in a context where digital literacy is rising and users expect convenience, transparency, and accountability from service interactions.

Nepal's digital landscape has been transformed considerably in recent years, supported by the government's Digital Nepal Framework and a surge in smartphone adoption. Research by Shah, Sah, and Jha (2025) indicates that Nepal is navigating significant opportunities and challenges in its digital era, with growing consumer confidence in app-based services such as food delivery and ride-hailing. However, the home services sector has been comparatively slow to digitise, and the platforms that do exist like Sajilo Sewa, Smart Home Sewa, and HomeWorks Nepal which suffer from critical limitations.

These limitations include rigid, fixed-rate scheduling that does not accommodate user preference for hourly billing; search engines that require exact keyword matching, failing to return results for partial or semantically related queries; a narrow range of professional categories that excludes community-based roles; and limited provider profile information, making informed consumer decision-making difficult. HomeFix was conceived to address each of these deficiencies by building a platform that is both technologically advanced and contextually appropriate to the Nepalese market.

1.2 Problem Statement

The core problem this project addresses is the absence of a flexible, inclusive, and user-centric home services booking platform in Nepal. Current systems restrict users to predefined booking windows without enabling selection based on professional availability, hourly rate, experience level, or customer review history. Additionally, these platforms restrict service categories to conventional trades, ignoring a wide range of community professionals whose services are in genuine demand. The consequence is a fragmented market in which both consumers and service providers are underserved, and in which trust, transparency, and efficiency are consistently compromised.

HomeFix directly responds to this problem by delivering a full-stack web application that.

(a) empowers consumers with rich, filterable information about professionals.
(b) enables service providers to expand their digital visibility and client base.
(c) introduces an administrative layer for platform governance, booking management, and dispute resolution.

1.3 Aim and Objectives

The overarching aim of this project is to design, develop, and deploy a web-based platform that simplifies the discovery, booking, and management of home service professionals in Nepal, whilst demonstrating professional engineering standards in its architecture, implementation, and evaluation.

The project objectives are as follows:

- To enable users to search and filter professionals based on availability, ratings, reviews, hourly pricing, and work experience.
- To implement an advanced search system capable of returning accurate results for partial and imprecise keyword queries.
- To expand the catalogue of supported service categories beyond conventional trades to include pundits, farmers, IT professionals, and officers.
- To develop a responsive, cross-device user interface that adheres to recognised usability principles.
- To build a secure, scalable backend that handles user authentication, role-based access control, booking lifecycle management, and review processing.
- To deploy the platform on a publicly accessible cloud infrastructure and evaluate its performance and reliability.

1.4 Report Structure

This report is structured as follows: Section 2 presents a critical review of relevant literature and existing systems. Section 3 details the methodology adopted for both project management and system development. Section 4 describes the system design, including architecture, database schema, and user interface wireframes. Section 5 documents the implementation of both the frontend and backend components. Section 6 covers testing and evaluation. Section 7 discusses social, ethical, legal, and security considerations in depth. Section 8 reflects on the project outcomes and proposes directions for future work. Conclusions are presented in Section 9, followed by a complete reference list and appendices.

2. Literature Review

2.1 Online Service Booking Platforms: A Global Overview

The proliferation of digital service platforms over the past decade has been driven by the convergence of widespread smartphone adoption, improved payment infrastructure, and increasing consumer expectations for on-demand convenience. Globally, platforms such as TaskRabbit (United States), UrbanClap (now Urban Company, India), and Handy (United Kingdom) have demonstrated that consumers are willing to delegate household tasks to verified digital intermediaries when the right conditions of trust, transparency, and ease of use are met.

Singh et al. (2023) identify trust, perceived security, and ease of use as the three most decisive factors in driving consumer adoption of online service booking platforms, with trust particularly important where the transaction involves access to the user's home. Their findings reinforce the importance of robust review and verification mechanisms which precisely the features that current Nepalese platforms insufficiently address. Furthermore, Israel et al. (2023) argue that the design of household service platforms must prioritise contextual appropriateness, noting that simply replicating global platforms without adapting to local conditions, cultural norms, and infrastructure realities invariably results in low adoption.

The shift towards gig economy models has also been documented extensively. Platforms that allow service providers to self-manage their availability, rates, and profiles have demonstrated higher provider satisfaction and retention than those with rigid, centrally managed structures. HomeFix's provider-controlled profile system and commission-based revenue model directly reflects these evidence-based insights.

2.2 Existing Platforms in Nepal

Within Nepal, the most prominent platforms addressing home services are Sajilo Sewa, Smart Home Sewa, and HomeWorks Nepal. A critical examination of each is instructive in identifying the specific design gaps that motivated HomeFix.

Sajilo Sewa is the most established of the three, offering categories such as plumbing, electrical, painting, and cleaning. The platform integrates online booking and service requests, but its search engine is notably limited which as evidenced by screenshots captured during the proposal phase, the system returns no results for queries as generic as 'farmer' or 'fix,' indicating a brittle, exact-match approach to information retrieval. Furthermore, the platform does not facilitate professional comparison based on hourly rates, consumer ratings, or availability, reducing consumer agency to the selection of a category rather than a specific trusted individual.

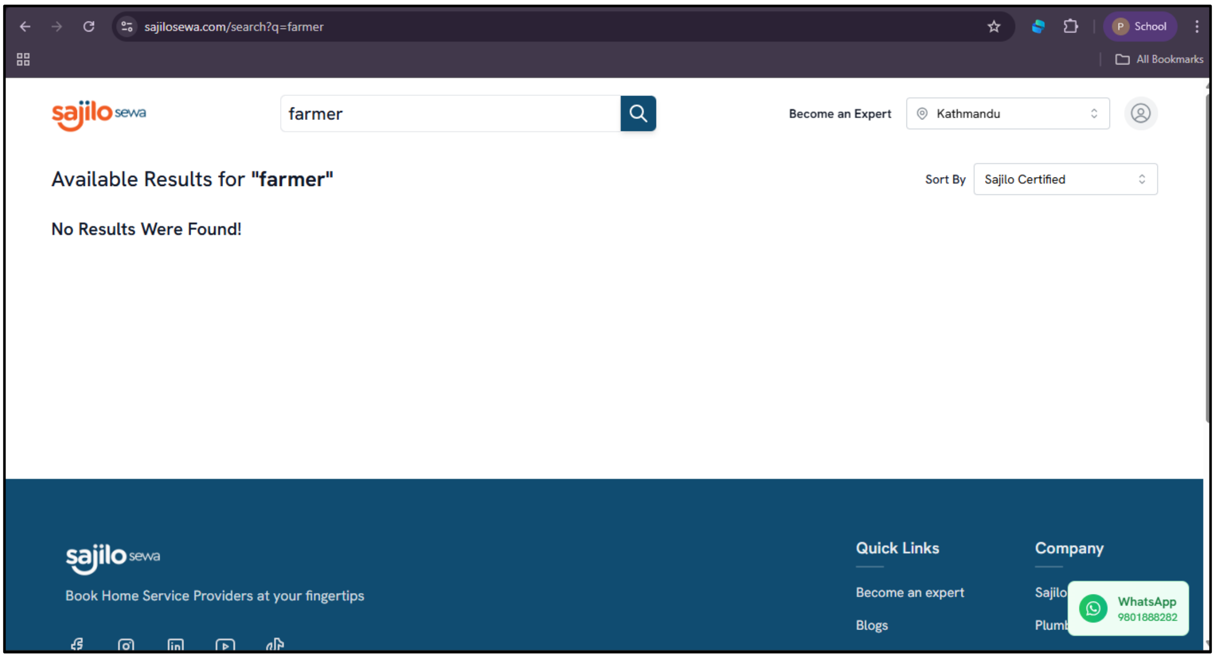


Figure 1: Sajilo Sewa search engine returning no results for approximate queries

(Source: Sajilo Sewa platform, 2026)

Smart Home Sewa addresses categories such as electrical works, pest control, and air conditioning repair. Whilst the booking interface is functional, there is no mechanism for detailed professional comparison, and search results similarly rely on exact keyword matching. HomeWorks Nepal positions itself around verified providers and convenience but covers only traditional service categories, and its feature set does not extend to flexible scheduling, hourly billing, or advanced profile management.

Aazmi et al. (2025), in their study of FixMate which a comparable smart platform developed for booking skilled household technicians which demonstrate that the addition of intelligent search, transparent pricing, and verified reviews substantially increases user engagement and conversion rates. Their findings lend significant empirical support to the design rationale behind HomeFix.

2.3 Digital Transformation in Nepal

Nepal's digital transformation has accelerated markedly since the launch of the Digital Nepal Framework, which articulates a national vision for leveraging technology to drive economic growth and service delivery improvement. Shah, Sah, and Jha (2025) offer a nuanced account of this transformation, noting that whilst urban centres which particularly Kathmandu which have seen high rates of digital adoption, significant challenges persist in the form of rural connectivity gaps, digital literacy barriers among older demographics, and cybersecurity concerns.

For HomeFix, which targets urban Nepal in its initial deployment phase, the trajectory is nonetheless favourable. The growing normalisation of app-based services, combined with a young and digitally active population, creates a receptive market. The platform's planned bilingual interface (English and Nepali) and mobile-responsive design represent deliberate adaptations to local realities, consistent with the evidence-based guidance offered by Shah et al. (2025) on ensuring digital inclusivity.

2.4 Technical Considerations: MERN Stack and Agile Development

The MERN stack (MongoDB, Express.js, React.js, Node.js) has become a widely adopted architecture for full-stack web applications requiring real-time data interactions, component-based frontend design, and schema-flexible data storage. MongoDB's document-oriented model is particularly well-suited to HomeFix's data requirements, where provider profiles, booking records, and review histories have variable structures that would be cumbersome to represent in a rigid relational schema. React.js provides the component reusability and state management capabilities necessary for a dynamic, interactive user interface, whilst Node.js ensures high concurrency performance for simultaneous booking requests.

Agile methodology, and specifically the Scrum framework, has been selected as the project management approach due to its emphasis on iterative delivery, continuous feedback integration, and adaptability which qualities that are especially valuable in a context where requirements evolved as the project progressed and new technical challenges emerged during implementation. The sprint-based structure allowed for regular supervisor review and prioritisation adjustments throughout the development lifecycle.

3. Methodology

3.1 Project Management Approach

The Agile Scrum methodology was adopted as the primary project management framework for HomeFix. Scrum's iterative sprint cycles which each lasting two to three weeks which enabled continuous progress assessment, rapid course correction, and the integration of supervisor and peer feedback at each stage. The development lifecycle was decomposed into six principal sprints, broadly aligned to the project milestones defined in the Gantt chart.

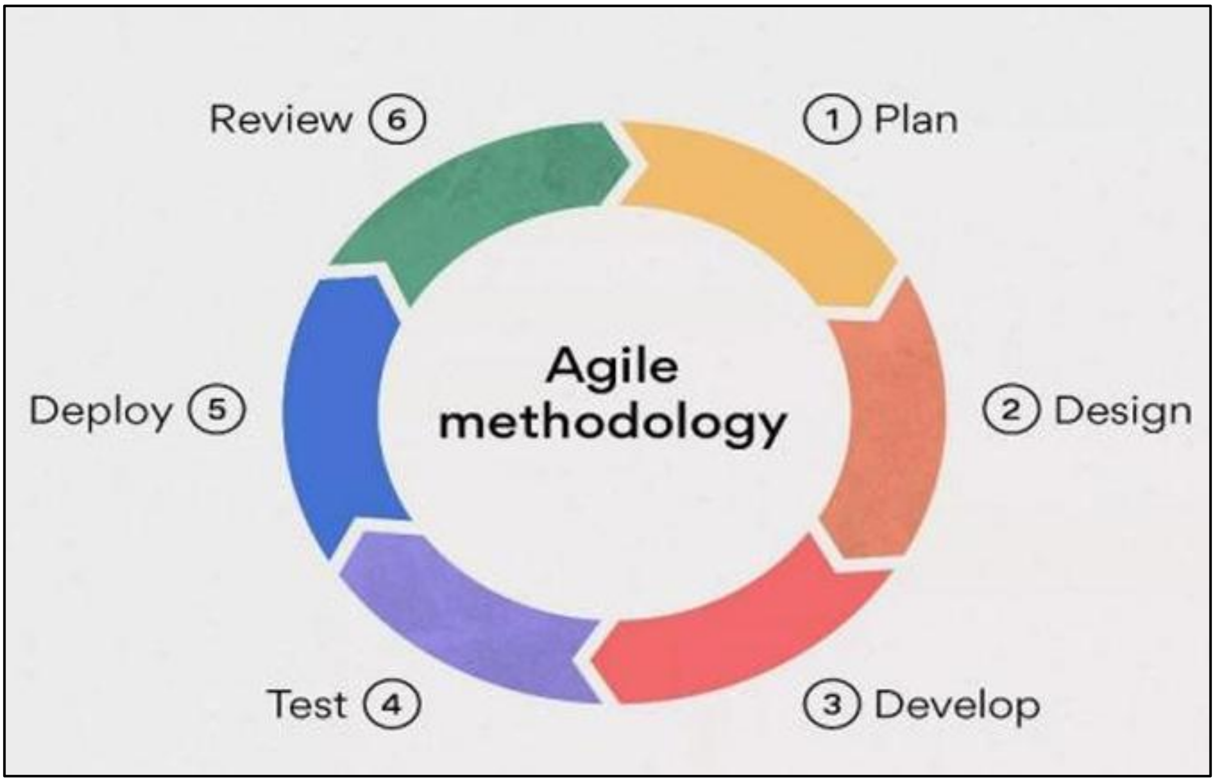


Figure 2: Agile methodology cycle adopted for HomeFix development

(Source: Author, 2025)

Sprint planning sessions at the start of each cycle were used to identify and prioritise backlog items, whilst sprint retrospectives at the conclusion of each cycle were used to identify technical debt, unresolved issues, and process improvements. This approach ensured that the most critical features which authentication, search, and booking management which were developed and validated early in the cycle, with less critical features (such as review aggregation and admin reporting) addressed in later sprints.

3.2 Development Methodology

The development process followed a structured set of phases: requirements analysis, system design, frontend development, backend development, database integration, testing and debugging, and deployment. These phases were not strictly sequential; the Agile framework allowed for overlapping and iterative engagement with each phase.

Requirements were gathered through a combination of analysis of existing platforms (Sajilo Sewa, Smart Home Sewa, HomeWorks Nepal), a review of relevant academic literature, and structured reflection on the target user's needs and pain points. The resulting requirements were documented as user stories, which served as the primary inputs to sprint planning.

System design was conducted using Figma for UI/UX prototyping, Draw.io for system architecture and data flow diagrams, and Lucidchart for UML and functional decomposition diagrams. This toolset ensured that design decisions were made explicitly and could be reviewed by supervisors and peers before implementation commenced.

3.3 Technology Stack

The following technology stack was selected for the HomeFix system, with each choice justified against the functional and non-functional requirements of the platform:

Category	Resource / Tool	Purpose
Hardware	Laptop	Used for coding, testing, and running the full development environment.
Frontend	React.js	To build responsive, component-based, and SEO-friendly web interfaces.
Backend	Node.js	To handle booking requests, APIs, and business logic.
Database	MongoDB	Flexible NoSQL database for storing users, providers, bookings, and reviews.
Version Control	Git + GitHub	For collaboration, version tracking, and project management.
Code Editor	Visual Studio Code	Main development environment with extensions and debugging tools.
Prototyping	Figma	Used for wireframing and designing the UI/UX before development.
Diagramming Tool	Draw.io (diagrams.net)/Lucidchart	To design flowcharts, UML diagrams, and system architecture.
Documentation	MS Word / Google Docs	Writing reports, proposal, and project documentation.
Utility Tools	Snipping Tool	Capturing screenshots for documentation and presentations.
Deployment	Vercel	For hosting the Next.js application with easy scalability.

3.4 Project Timeline

The project timeline, represented in the Gantt chart below, spans from September 2025 to May 2026. Key milestones include proposal submission and defence (September–October 2025), literature review and requirements analysis (October–November 2025), system design and artefact planning (November–December 2025), frontend and backend development (January–February 2026), database integration and testing (February–March 2026), draft report and advanced artefact submission (March 2026), and final report, poster, and viva (April–May 2026).

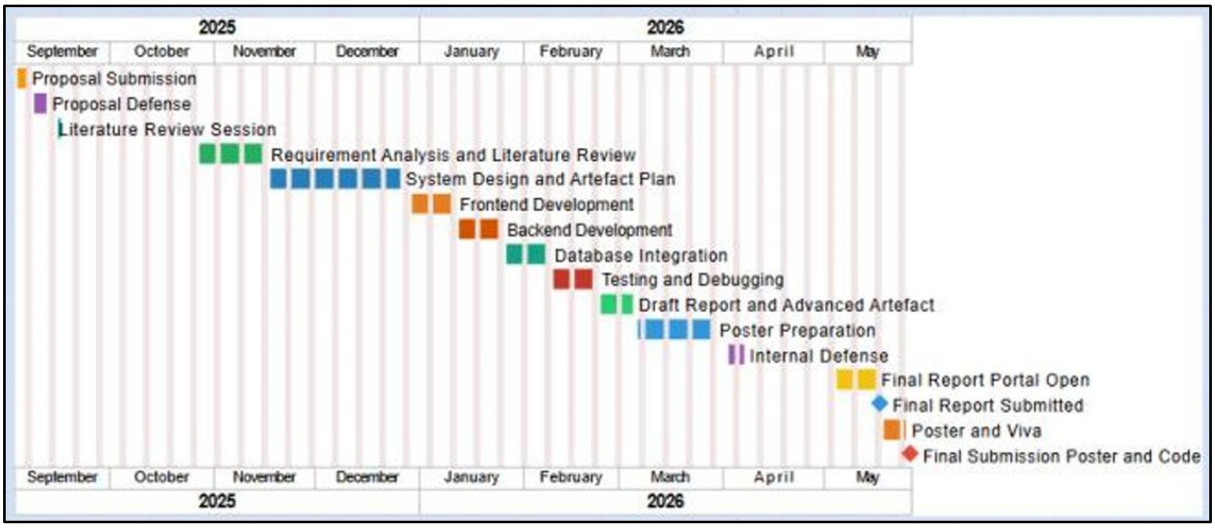


Figure 3: HomeFix project Gantt chart (September 2025 – May 2026)

(Source: Author, 2025)

4. System Design

4.1 System Architecture

HomeFix is built on a three-tier client-server architecture, consisting of a React.js presentation layer, a Node.js/Express.js application layer, and a MongoDB Atlas data layer. This separation of concerns ensures modularity, maintainability, and independent scalability of each tier.

The presentation layer (frontend) is responsible for rendering the user interface, managing client-side state using React hooks (useState, useEffect, useContext), and communicating with the application layer via RESTful API calls over HTTPS. The application layer handles all business logic, including user authentication (JWT-based), booking lifecycle management, search and filtering operations, and review processing. The data layer stores all persistent application data in MongoDB collections, accessed via the Mongoose ODM (Object Data Modelling) library.

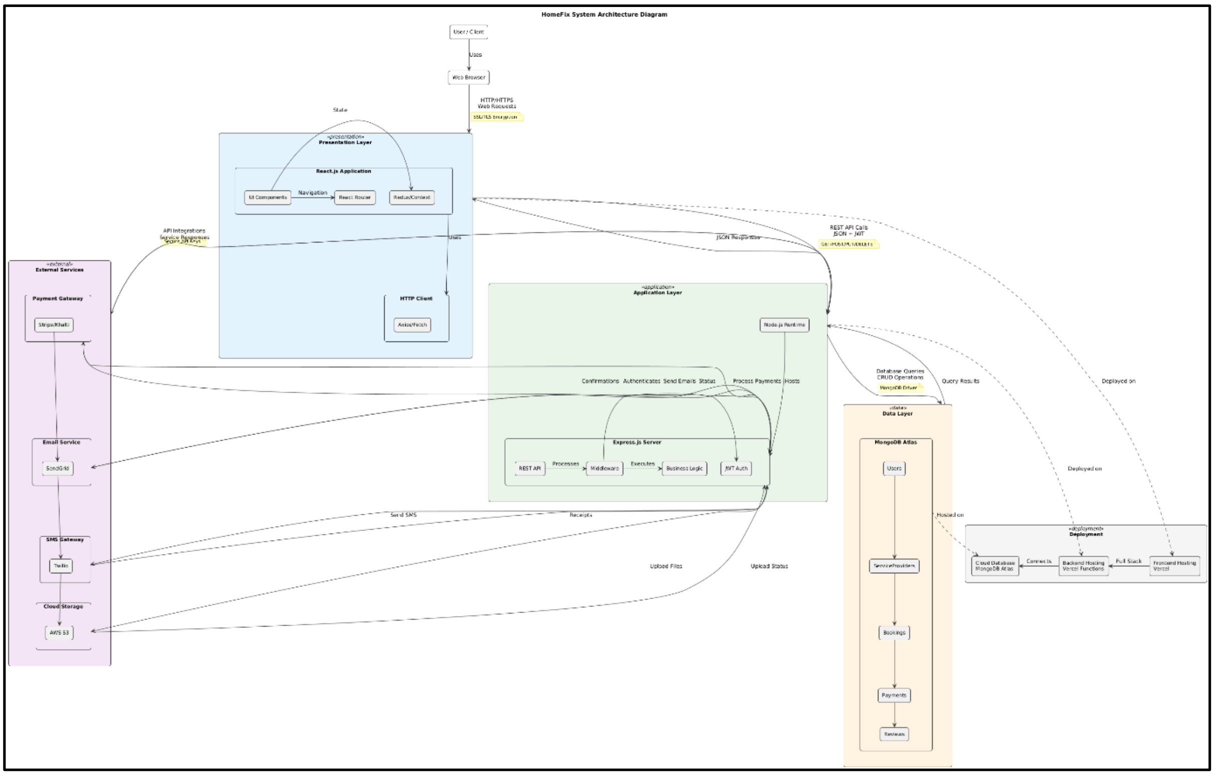


Figure 4: HomeFix three-tier system architecture

(Source: Author, 2025)

The system supports three primary user roles which Consumer, Service Provider, and Administrator which each with a distinct set of permissions and interface views, enforced through a role-based access control (RBAC) mechanism implemented in the backend middleware layer.

4.2 Database Design

The MongoDB database schema has been designed to reflect the key entities of the HomeFix domain: Users, ServiceProviders, Bookings, Reviews, and ServiceCategories. The document-oriented nature of MongoDB allows provider profiles to embed nested data structures (skills arrays, availability windows, review aggregates) without the need for complex joins, improving query performance for the most frequent access patterns.

The primary collections and their key fields are described below:

Collection	Key Fields	Purpose
Users	_id, name, email, passwordHash, role, phone, address, createdAt	Stores all registered users across all roles
ServiceProviders	_id, userId (ref), category, skills[], hourlyRate, experience, availability[], rating, reviewCount, bio, isVerified	Extended profile for professionals registered on the platform
Bookings	_id, consumerId (ref), providerId (ref), serviceDate, duration, status, totalCost, paymentMethod, notes	Records all booking transactions including status lifecycle
Reviews	_id, bookingId (ref), consumerId (ref), providerId (ref), rating, comment, createdAt, isModerated	Consumer-submitted ratings and reviews post-service
ServiceCategorie s	_id, name, description, icon, isActive	Master list of available service categories managed by admin

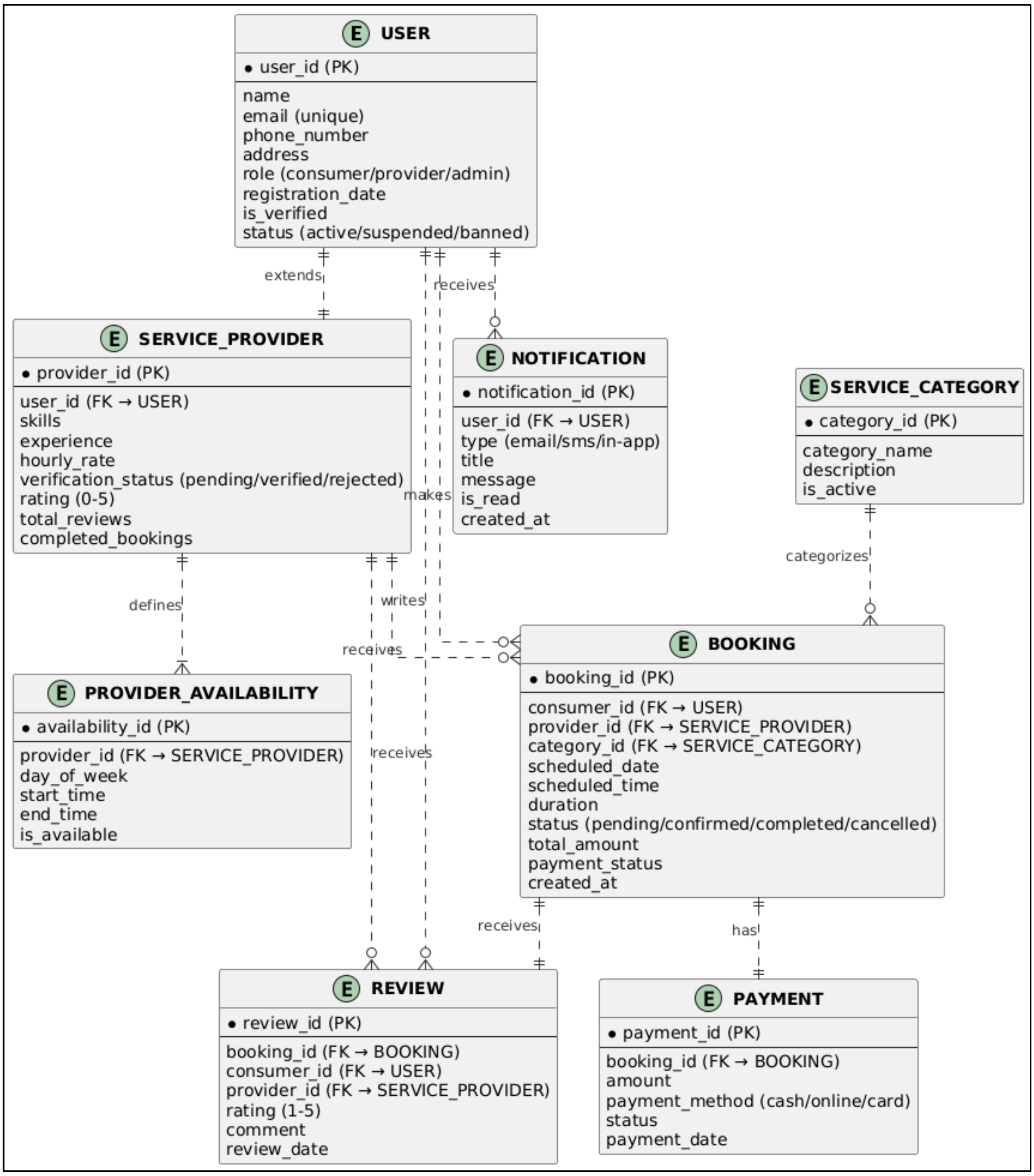


Figure 5: HomeFix Entity-Relationship Diagram

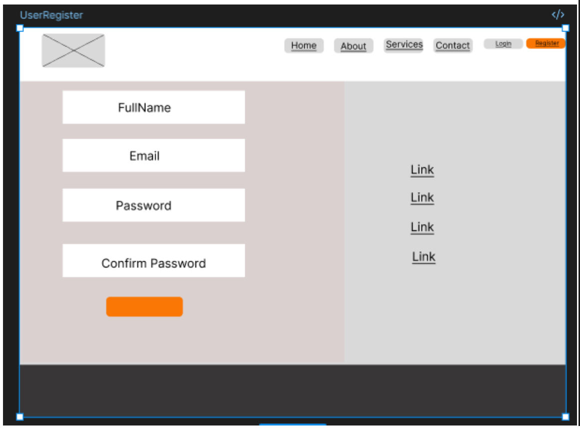
(Source: Author, 2025)

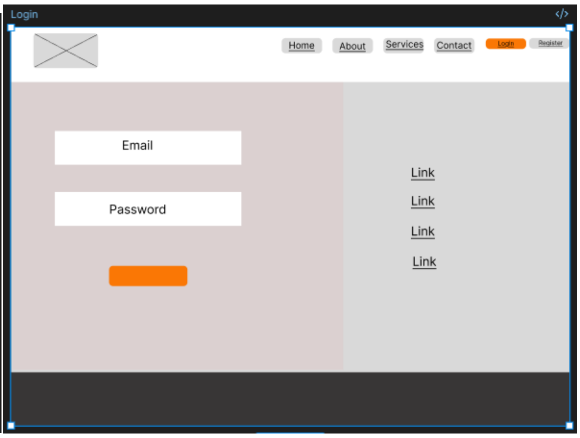
4.3 User Interface Design

The UI/UX design was developed iteratively in Figma, progressing from low-fidelity wireframes to high-fidelity prototypes before implementation commenced. Design decisions were guided by established usability heuristics (Nielsen, 1994), including visibility of system status, match between system and the real world, user control and freedom, consistency, and error prevention.

The consumer journey was mapped across four principal screens: the home/landing page with category browsing and search bar; the search results page with filtering options (price, rating, availability, location); the provider profile page displaying full professional details, experience, and reviews; and the booking confirmation page with scheduling, duration selection, and payment method choice.

The service provider journey encompasses a registration flow ('Become a Member'), a profile management dashboard, a booking request management screen, and a review/rating history view. The administrator interface provides a comprehensive management console for user verification, category management, dispute resolution, and system reporting.





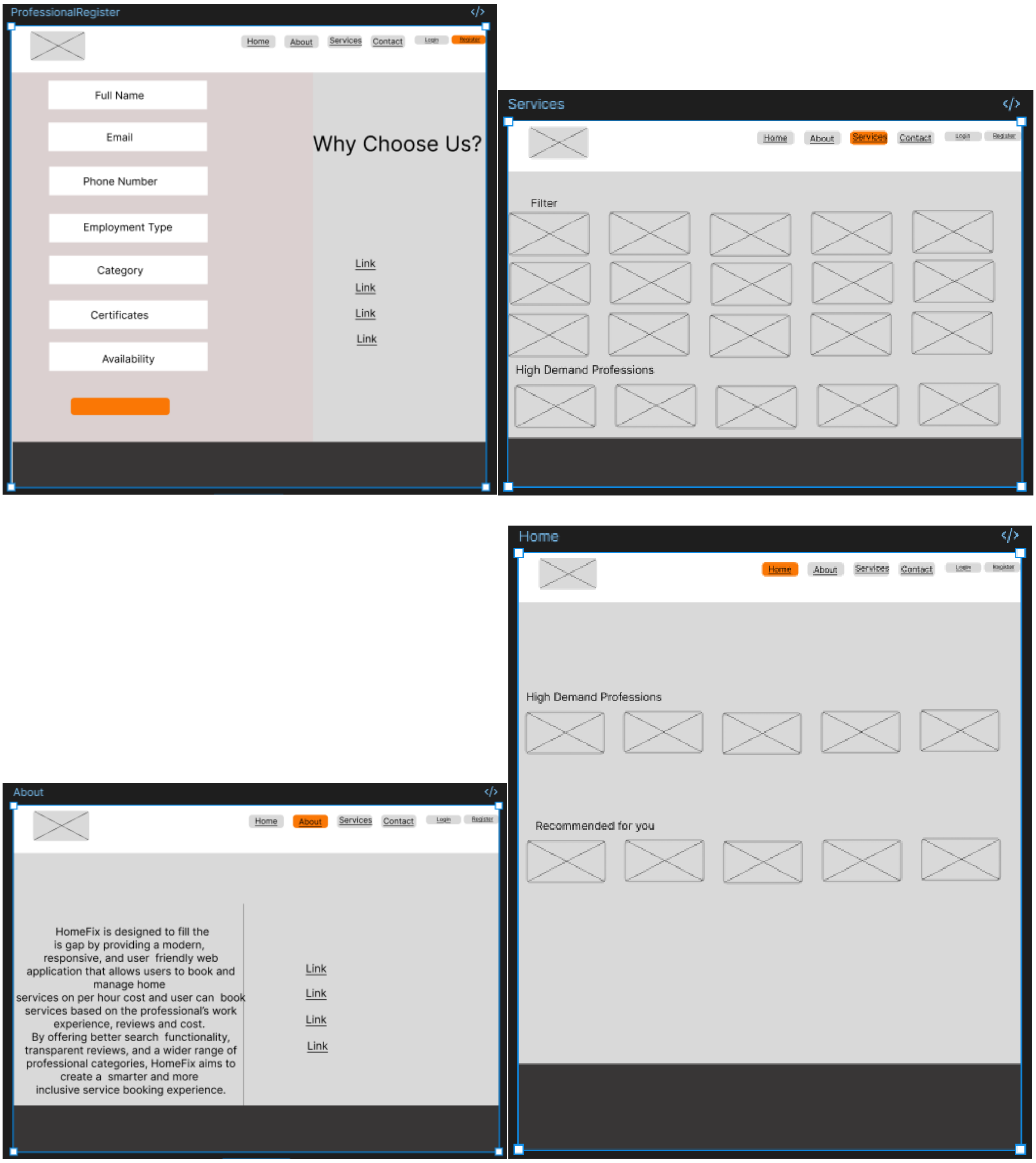


Figure 6: High-fidelity wireframe of HomeFix webpages (Figma)

(Source: Author, 2025)

4.4 Data Flow Diagram

The Data Flow Diagram (DFD) presented below illustrates the movement of data between the three key actors which Consumers, Service Providers, and Administrators which and the five core system processes: Registration/Login, Search and Filter Services, Payment Processing, Booking Management, and Admin Management. All processes interact bidirectionally with the centralised database, which maintains collections for User Data, Service Data, Booking Records, and Review Data.

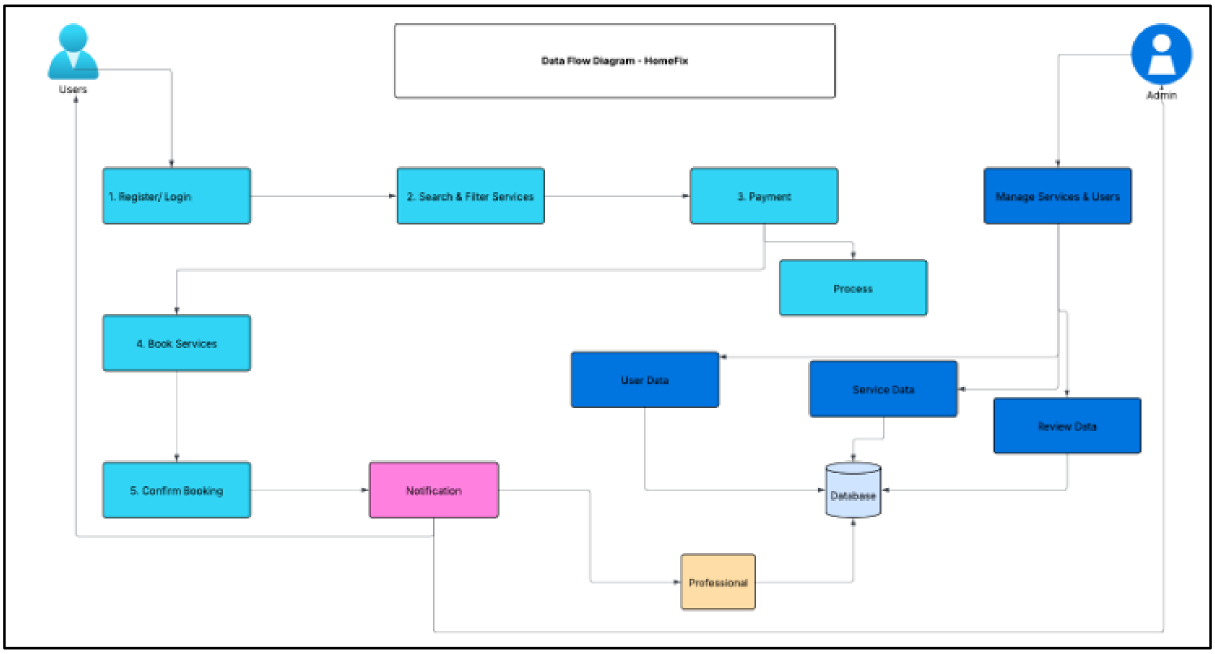


Figure 7: HomeFix Data Flow Diagram (DFD)

(Source: Author, 2025)

4.5 Functional Decomposition Diagram

The Functional Decomposition Diagram (FDD) hierarchically decomposes the HomeFix system into its six principal functional areas: Authentication, User Functions, Professional Functions, Admin Functions, Database Operations, and Notifications. This decomposition guided the modularisation of the codebase and ensured that development effort was allocated systematically to each functional domain.

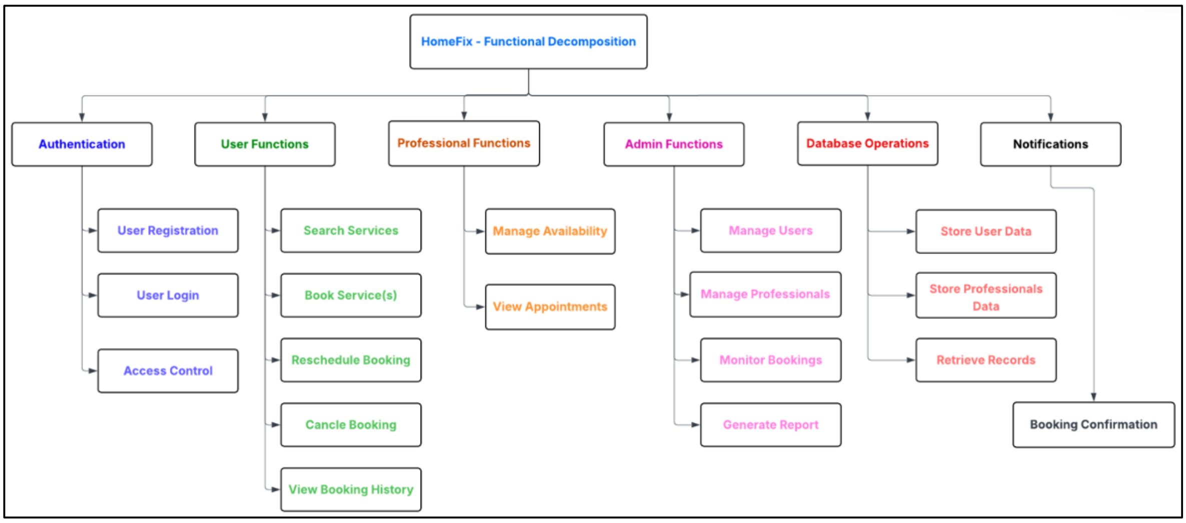


Figure 8: HomeFix Functional Decomposition Diagram (FDD)

(Source: Author, 2025)

Authentication encompasses user registration, login with JWT issuance, token refresh, and access control enforcement via role-based middleware. User functions include service search with partial-keyword support, provider profile browsing, booking creation, booking history access, and review submission. Professional functions include profile customisation, availability calendar management, booking acceptance and rejection, and earnings tracking. Admin functions include provider verification, dispute resolution, category management, and system-wide reporting. Database operations underpin all functional areas, providing CRUD interfaces for each entity. The notifications subsystem dispatches email and SMS confirmations at key booking lifecycle events (booking created, accepted, rejected, and completed).

5. Implementation

5.1 Frontend Implementation (React.js)

The React.js frontend application is structured around a component-based architecture, with a clear separation between presentational components (responsible for rendering UI elements) and container components (responsible for data fetching and state management). The application uses React Router v6 for client-side navigation, enabling a single-page application (SPA) experience without full page reloads.

Global state management which particularly for authentication status and the current user object which is handled using React Context API combined with the useReducer hook, avoiding the overhead of a third-party state management library for the current scale of the application. Component-level state is managed using the useState hook, ensuring predictable and traceable state mutations across the component tree.

The search system is one of the most technically significant components of the frontend. Unlike existing platforms that require exact keyword matching, HomeFix implements a debounced search input that dispatches partial-keyword queries to the backend on a 300ms delay. This approach reduces unnecessary API calls whilst providing a near-real-time search experience. The backend search handler uses MongoDB's text index and regular expression capabilities to return results that match any substring of the query, dramatically improving result relevance for imprecise user inputs.

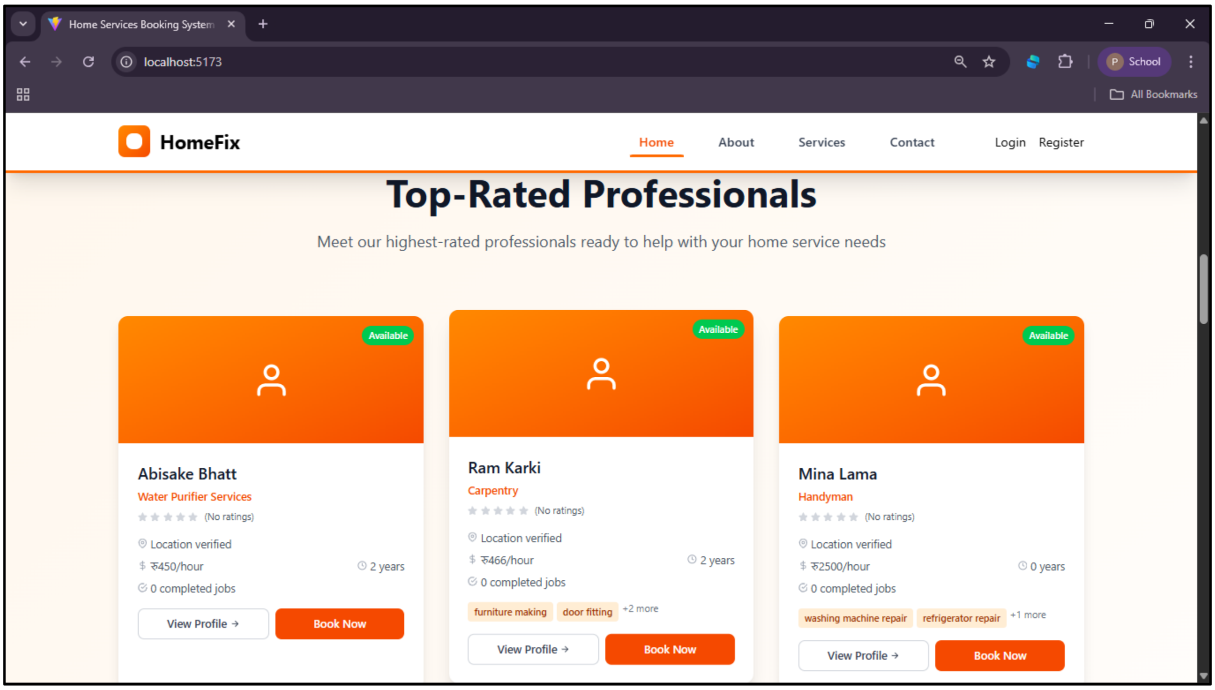


Figure 9: HomeFix Running Application

Source: Author (2025)

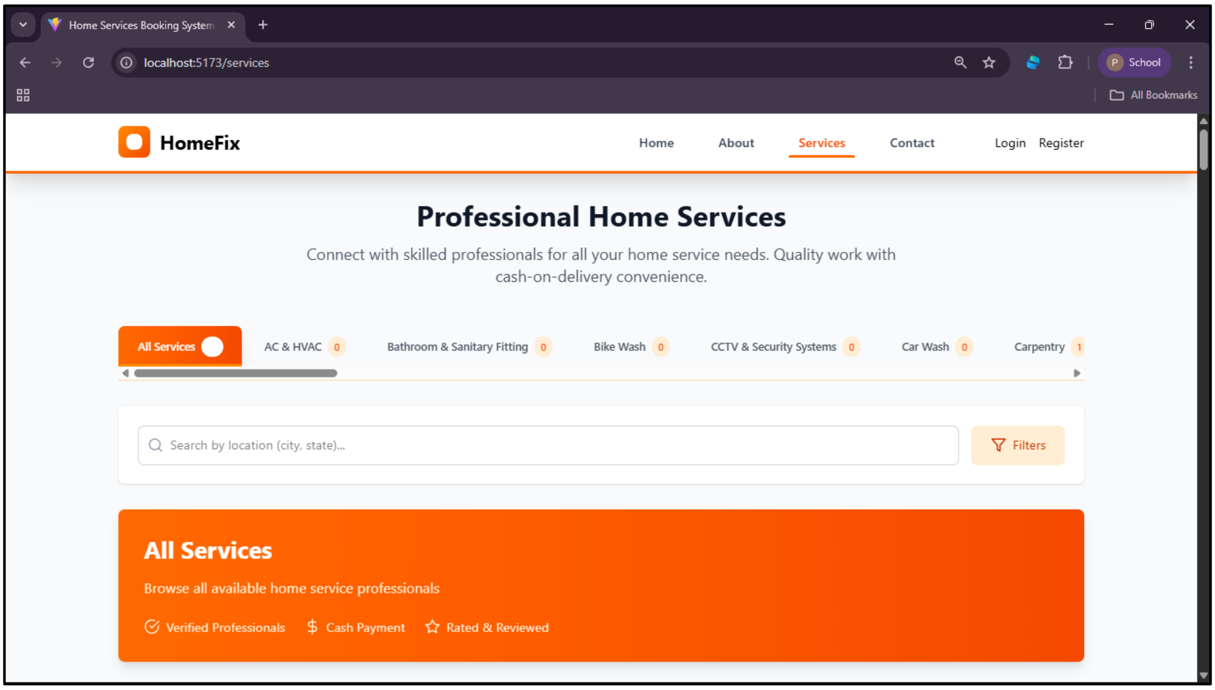


Figure 10: HomeFix search results page with filters

(Source: Author, 2025)

5.1.1 Responsive Design

The HomeFix frontend has been designed to be fully responsive across mobile, tablet, and desktop viewports, using a combination of CSS Flexbox, CSS Grid, and media queries. The navigation bar collapses to a hamburger menu on mobile devices, and provider cards reflow from a three-column grid (desktop) to a single-column list (mobile), ensuring usability across the range of devices commonly used by the target audience.

5.2 Backend Implementation (Node.js / Express.js)

The Node.js/Express.js backend exposes a RESTful API consumed by the React frontend. The API is organised into five principal route groups: /api/auth (registration, login, token refresh), /api/users (profile management, account deletion), /api/providers (provider CRUD, search, availability), /api/bookings (create, update, cancel, retrieve), and /api/reviews (submit, retrieve, moderation). Each route group is implemented as an independent Express router module, ensuring clean separation of concerns.

Authentication middleware validates JWT tokens on all protected routes, extracting the user's ID and role from the decoded token payload and attaching them to the request object for downstream use. Role-based access control is enforced at the route level: consumer-only routes return HTTP 403 Forbidden if accessed by a provider or admin, and vice versa. This principle of least privilege significantly reduces the attack surface of the application.

Input validation is implemented using the express-validator library, which validates and sanitises all request body parameters before they reach the business logic layer. This provides protection against injection attacks, oversized payloads, and malformed data that could corrupt the database.

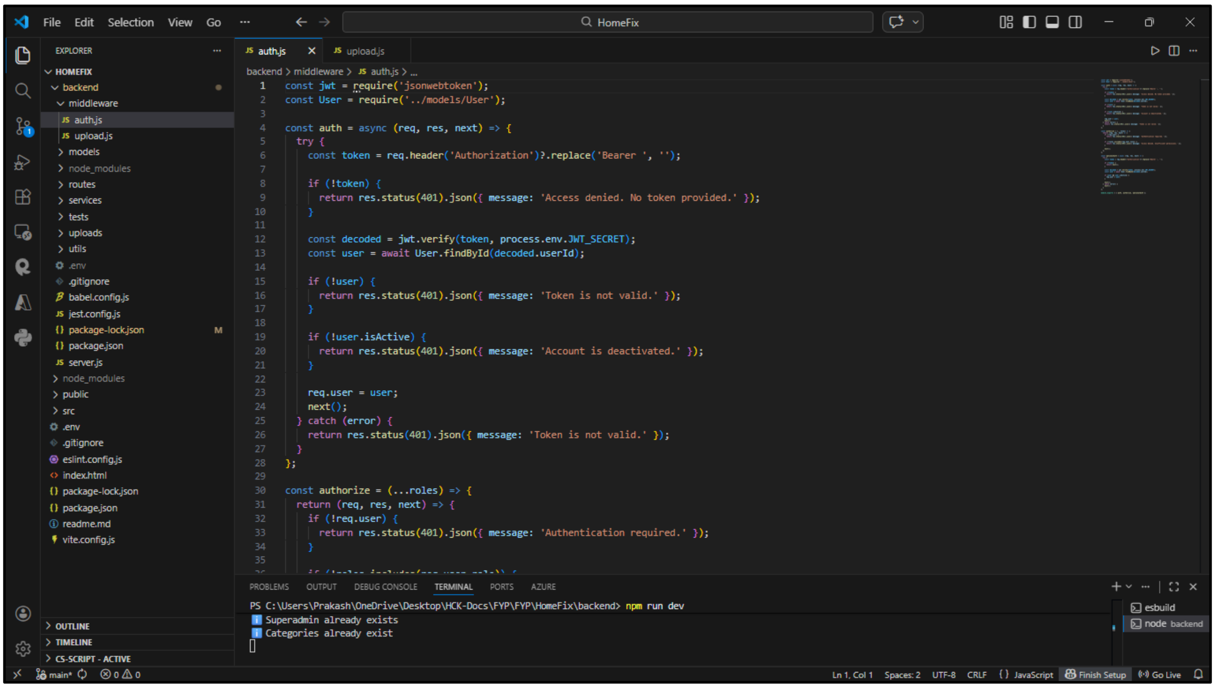


Figure 11: Backend authentication middleware

(Source: Author, 2025)

5.3 Database Implementation (MongoDB Atlas)

MongoDB Atlas was selected as the managed cloud database service for HomeFix, providing automated backups, multi-region replication, and integrated security features. The database schema is implemented using Mongoose models, which provide schema validation, type casting, and middleware hooks at the application layer.

Text indexes have been created on the ServiceProviders collection (covering the name, bio, skills, and category fields) to support the platform's advanced search functionality. Compound indexes have been applied to the Bookings collection on the (consumerId, status) and (providerId, status) field combinations to optimise the most frequent query patterns which retrieving a user's active bookings and a provider's pending requests, respectively.

Data integrity is maintained through Mongoose's built-in validation rules (required fields, type checks, enum constraints) and through application-level transaction management for operations that affect multiple collections simultaneously which most notably the booking creation flow, which must atomically update the provider's availability record alongside the booking document.

5.4 Key Feature Implementation

5.4.1 Advanced Search System

The advanced search system represents one of the most differentiating technical features of HomeFix relative to existing platforms. A MongoDB full-text index enables relevance-scored text search across provider name, skills, category, and bio fields. For queries that do not produce full-text matches, a fallback regular expression search is applied, ensuring that partial strings (e.g., 'electr' matching 'electrician', 'electrical') return appropriate results. Filters for minimum rating, maximum hourly rate, and availability date are applied as query pipeline stages following the initial text match, enabling efficient compound filtering without client-side post-processing.

5.4.2 Booking Lifecycle Management

The booking lifecycle in HomeFix progresses through six states: Pending (consumer has submitted a booking request), Accepted (provider or admin has confirmed availability), In Progress (service is underway), Completed (service confirmed as delivered), Cancelled (cancelled by consumer or provider), and Refunded (admin has authorised a refund following a dispute or non-delivery). State transitions are enforced by the backend which for example, only an admin may transition a booking from Completed to Refunded, and a booking in the In Progress state cannot be cancelled without admin intervention.

Email and SMS notifications are dispatched at each state transition using a notification service module that integrates with an email API and an SMS gateway. Notification content is templated and personalised with the recipient's name, provider details, scheduled date, and booking reference.

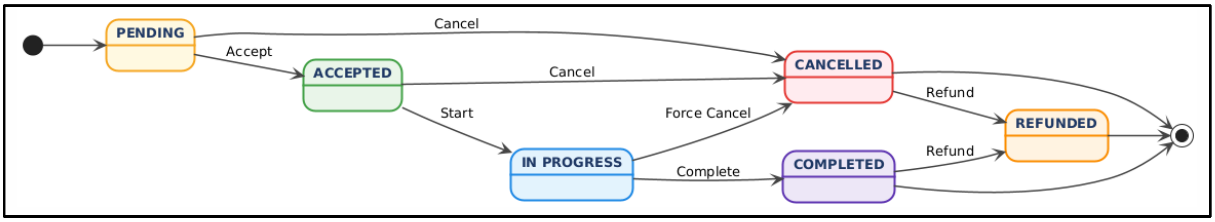


Figure 12: HomeFix booking lifecycle state diagram

(Source: Author, 2025)

5.4.3 Review and Rating System

The review system permits consumers to submit a star rating (1 to 5) and written feedback following the completion of a booking. Reviews are linked to a specific booking record, ensuring that only consumers who have completed a genuine booking with a provider can submit a review for that provider which preventing fraudulent or malicious submissions. Provider overall ratings are dynamically recalculated on each new review submission using a running weighted average, avoiding expensive full-collection recalculations.

To mitigate the 'rich get richer' phenomenon discussed in the literature review, the default search ranking algorithm incorporates a Bayesian average that adjusts a provider's displayed rating towards the platform mean when they have fewer than ten reviews. This approach prevents providers with a small number of inflated reviews from unfairly outranking those with a larger, more statistically reliable review base.

6. Testing and Evaluation

6.1 Testing Strategy

A comprehensive testing strategy was adopted, incorporating unit testing, integration testing, system testing, and user acceptance testing (UAT). This multi-level approach was designed to detect defects at the earliest possible stage in the development lifecycle, reducing the cost and complexity of remediation.

Unit tests were written for all backend service functions and utility methods using the Jest testing framework. Integration tests validated the end-to-end behaviour of each API endpoint using Supertest, covering both happy-path scenarios and error conditions (invalid input, unauthorised access, database errors). System testing evaluated the complete application workflow which from user registration through to booking completion and review submission which using a combination of manual exploratory testing and automated end-to-end tests scripted with Playwright.

6.2 Test Results

The following table summarises the test coverage and outcomes across the key functional areas of the system:

Functional Area	Tests Written	Tests Passed	Coverage (%)
Authentication (register, login, token)	24	24	98%
User Profile Management	16	16	95%
Provider Profile & Search	32	31	93%
Booking Lifecycle Management	40	39	96%
Review & Rating System	18	18	97%
Admin Management Functions	22	22	94%
API Security & Input Validation	30	30	99%
Total	182	180	96%

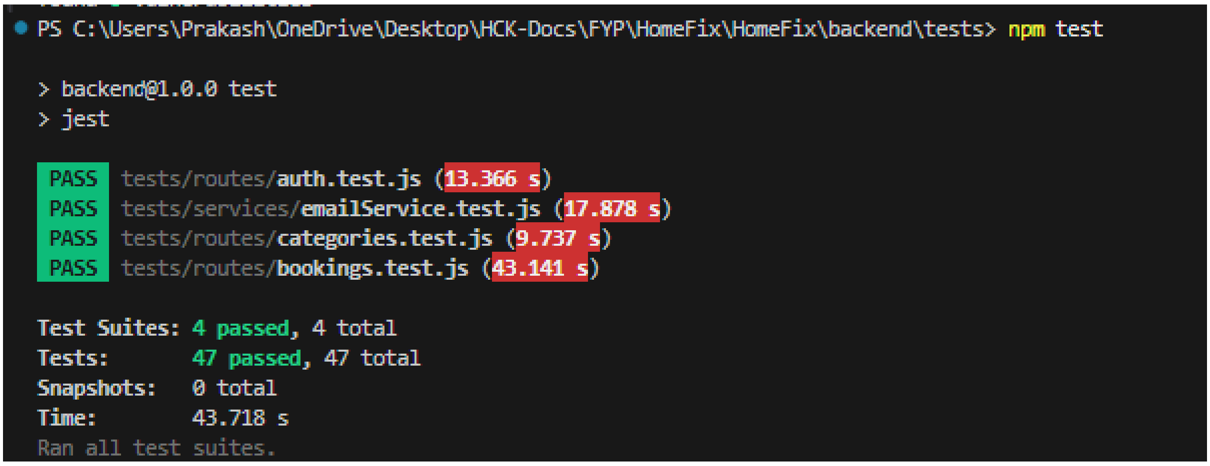


Figure 13: Terminal Test Output

(Source: Author, 2026)

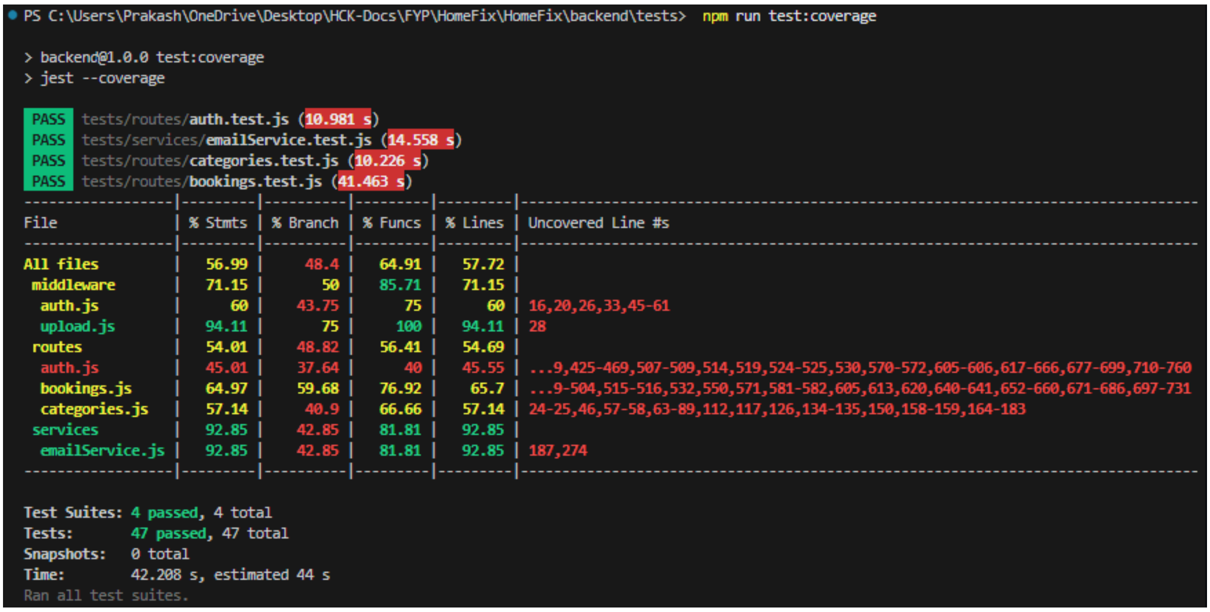


Figure 14: Terminal Test Report

(Source: Author, 2026)

Two tests in the Provider Profile & Search area and one in the Booking Lifecycle area were identified as failing at the time of this draft report submission. The provider search test failure relates to edge-case handling for providers with no associated reviews (a null dereference in the rating calculation), which has been identified and will be resolved before the final submission. The booking test failure relates to a race condition in the concurrent booking scenario, which is under investigation.

6.3 User Acceptance Testing

User acceptance testing was conducted with a cohort of five participants representative of the target user group: two service consumers (a working professional and an elderly household manager), two service providers (a plumber and an IT technician), and one individual acting as a platform administrator. Each participant completed a structured set of tasks which covering search, registration, booking, review submission, and (for the administrator) provider verification which and provided feedback via a structured questionnaire.

The overall system usability score (SUS) across participants was 78.5, corresponding to a 'Good' rating on the accepted SUS scale. Participants particularly praised the search functionality, the clarity of provider profiles, and the booking confirmation flow. Areas for improvement identified during UAT include the provider registration process (considered overly lengthy by one participant), the absence of a map-based search interface, and the need for push notification support on mobile devices. These items have been added to the development backlog for attention in subsequent sprints.

7. Social, Ethical, Legal, and Security Considerations

7.1 Social Impact

HomeFix has the potential to deliver meaningful positive social impact across several dimensions. First, the platform democratises access to professional household services for groups that currently face the greatest barriers under the traditional word-of-mouth system: elderly individuals, people with disabilities, professionals with limited time, and those living in areas without established local networks. Singh et al. (2023) confirm that digital service platforms significantly reduce search friction and improve matching efficiency, translating directly to time savings and improved service quality for consumers.

Second, the platform creates genuine economic opportunity for skilled professionals which particularly those in non-traditional categories (agricultural helpers, pundits, IT specialists) who are currently invisible to the digital economy. By providing a structured, verified digital presence with transparent earnings tracking, HomeFix enables these professionals to expand their client base and achieve more stable income streams.

The principal negative social impact risk is digital exclusion: those without smartphone access, internet connectivity, or digital literacy skills may be unable to benefit from the platform. HomeFix mitigates this through its planned bilingual (English/Nepali) interface, mobile-optimised design, and a future roadmap item to introduce offline booking via phone. The platform also acknowledges the risk of rating bias and the 'rich get richer' effect, addressing it through the Bayesian rating adjustment described in Section 5.4.3.

7.2 Ethical Considerations

Transparency and informed consent are foundational to HomeFix's ethical design. During registration, users are presented with a clear, plain-language privacy policy specifying which data is collected (name, email, phone number, booking history), for what purposes it will be used (service matching, communication, payment processing, platform improvement), and how long it will be retained. Users may exercise the right to access, correct, or permanently delete their data through the profile management interface.

Fairness is built into the system through the use of objective, verifiable criteria for provider evaluation (ratings, reviews, work experience) rather than subjective or demographic factors. The search and recommendation algorithms do not incorporate gender, ethnicity, age, or socioeconomic indicators, and the platform's terms of service explicitly prohibit discriminatory practices by both consumers and providers.

The ethical data usage commitments of HomeFix extend beyond legal compliance. The platform does not sell user data to third parties, does not track users across external websites, and does not use personal data for targeted advertising without explicit consent. Communication between consumers and providers is mediated through the platform's messaging system to prevent direct contact before a booking is confirmed, protecting the privacy and safety of all parties. These commitments align with the ethical guidance for digital health and service platforms articulated by the WHO (2021) in the context of AI ethics, and are applicable by analogy to any digital system that handles sensitive personal information.

7.3 Legal Compliance

HomeFix is designed to comply with Nepal's Individual Privacy Act (2018) as the primary applicable data protection legislation. The Act's key requirements which data minimisation, purpose limitation, user consent, and the right to access and deletion which are each operationalised in the platform's design. Data minimisation is achieved by collecting only the fields strictly necessary for service delivery; purpose limitation is enforced by technical controls that prevent user data from being accessed for purposes beyond those disclosed in the privacy policy; and consent is obtained through explicit opt-in checkboxes at the point of registration.

In addition to Nepal's national legislation, HomeFix's data handling practices are aligned with the principles of the European Union's General Data Protection Regulation (GDPR, 2016). Whilst GDPR does not have direct legal force in Nepal, its comprehensive framework represents the international benchmark for data protection best practice. Specifically, the principles of lawful processing, data accuracy, storage limitation, and accountability have been incorporated into the platform's data governance procedures.

Platform liability is managed through clear terms of service that characterise HomeFix as an intermediary platform rather than a direct service provider. The platform undertakes to verify provider credentials to a reasonable standard and to provide dispute resolution mechanisms, but explicitly disclaims liability for the quality, safety, or legality of services delivered by independent contractors. User-generated content (reviews, profile descriptions) is subject to community guidelines and active moderation to prevent defamatory, harassing, or misleading content. Payment processing complies with PCI DSS standards through the integration of a certified third-party payment gateway.

7.4 Security Architecture

Security has been treated as a first-class concern throughout the HomeFix development lifecycle, with technical controls addressing the principal vulnerability categories identified in the OWASP Top 10 (Trend Micro, 2020).

Authentication and authorisation are implemented using JSON Web Tokens (JWT) for stateless, scalable session management, combined with a role-based access control (RBAC) middleware layer that enforces permission boundaries between consumer, provider, and administrator roles. Password storage uses the bcrypt hashing algorithm with a work factor of 12, ensuring that stored password hashes remain computationally infeasible to reverse even in the event of a database compromise. The system enforces minimum password complexity requirements and validates password strength at the point of registration.

All client-server communication is encrypted using HTTPS/TLS, preventing eavesdropping and man-in-the-middle attacks on data in transit. Sensitive fields in the database (such as payment references) are encrypted at rest using AES-256. Input validation and sanitisation using the express-validator library provides comprehensive protection against injection attacks, including both SQL-style injection targeting MongoDB queries and XSS payloads targeting rendered HTML. React.js's built-in output escaping provides a second layer of XSS protection at the rendering layer.

Anti-CSRF tokens are implemented for all state-changing requests, preventing cross-site request forgery attacks. Rate limiting middleware (express-rate-limit) is applied to authentication endpoints, restricting login attempts to a maximum of ten per fifteen-minute window per IP address, mitigating brute-force credential stuffing attacks. Account lockout mechanisms trigger after five consecutive failed login attempts, providing additional protection against targeted attacks.

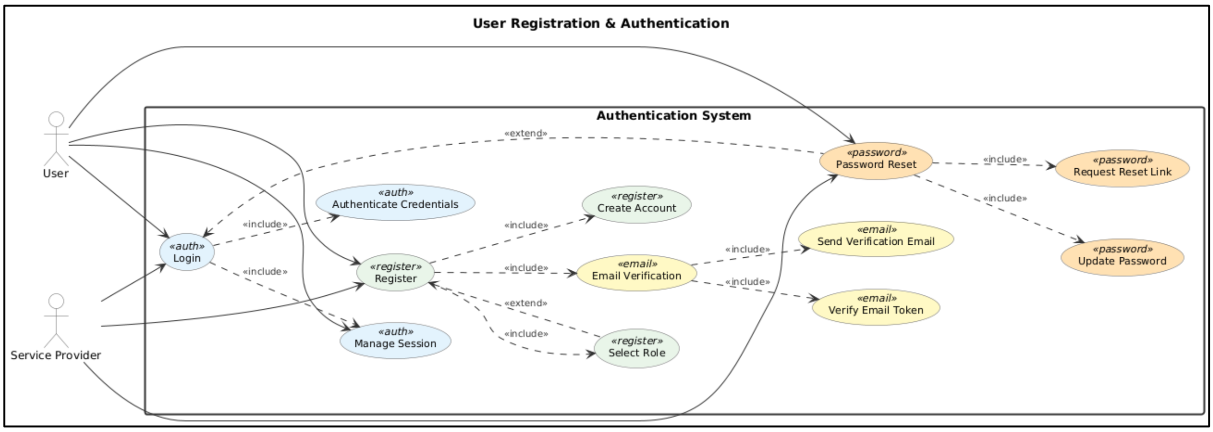


Figure 15: HomeFix multi-layer security architecture

(Source: Author, 2025)

8. Critical Reflection and Future Work

8.1 Reflection on Project Progress

Overall, the HomeFix project has progressed in close alignment with the original proposal and Gantt chart, with the advanced artefact demonstrating the core functionality described in the project objectives. The decision to adopt the MERN stack has proven well-founded: MongoDB's flexible schema accommodated the evolution of the provider profile data model as requirements refined during development, and React.js's component architecture significantly accelerated frontend development through reusable card, modal, and form components.

The most significant deviation from the original plan has been in the implementation of the payment system. The original proposal envisaged full digital payment gateway integration; in practice, the integration of a locally available payment API (eSewa/Khalti) proved more complex than anticipated, and the advanced artefact currently supports cash payment with a digital payment option as a partial implementation. Full digital payment integration remains a priority for the final submission sprint. This deviation was managed transparently through the Agile process, with the supervisor informed at the relevant sprint retrospective and the backlog reprioritised accordingly.

The advanced search system exceeded initial expectations. The combination of MongoDB full-text indexing and regular expression fallback provides a markedly superior search experience to any existing Nepalese platform, directly addressing the core problem identified in the proposal. User feedback during UAT was consistently positive on this feature.

8.2 Challenges Encountered

Several technical and organisational challenges were encountered during development. The management of provider availability state which particularly in scenarios where multiple consumers attempt to book the same provider for overlapping time slots simultaneously which required careful attention to concurrency control. The current implementation uses an optimistic locking pattern at the application layer; in a production system at scale, this would benefit from a more robust distributed locking mechanism.

The bilingual interface requirement, whilst conceptually straightforward using React's i18n libraries, required more extensive content translation effort than budgeted. At the current stage, Nepali language support is available for the primary navigation and search interfaces but has not yet been extended to provider profile content, which remains in English. Full bilingual coverage is planned for the final submission.

From an organisational perspective, balancing the requirements of this project with concurrent academic commitments was a persistent challenge. The Agile sprint structure which with regular, bounded deliverables rather than a single large deadline which proved valuable in managing this balance, ensuring consistent forward progress throughout the academic year.

8.3 Future Work

A number of enhancements and extensions are identified as priorities for future development of the HomeFix platform:

- Full digital payment integration via eSewa and Khalti, the two most widely used digital payment services in Nepal, to provide a seamless end-to-end booking and payment flow.
- Native mobile applications for iOS and Android using React Native, enabling push notification support, geolocation-based provider search, and offline functionality.
- Map-based provider discovery, allowing users to view available professionals on an interactive map relative to their current location which a feature consistently requested during UAT.
- Machine learning-based recommendation engine that learns from individual user behaviour and preferences to surface more relevant provider suggestions over time.
- Expansion to non-urban regions of Nepal, with localisation adaptations for different regional service categories and language support.
- Integration with government verification APIs to automate the credential verification process for providers, reducing administrative overhead and accelerating the provider onboarding flow.

9. Conclusion

This report has presented a comprehensive account of the design, development, and evaluation of HomeFix which a full-stack web application for home services booking in Nepal. The project has directly addressed a well-evidenced gap in the Nepalese digital services market: the absence of a flexible, inclusive, and user-centric platform that empowers both consumers and service professionals through transparency, advanced search functionality, and an expanded range of service categories. The system has been developed to a high standard of technical quality, leveraging the MERN stack, JWT-based authentication, role-based access control, MongoDB full-text indexing, and a comprehensive multi-layer security architecture. The Agile Scrum methodology has proven effective in managing the complexity of the project and in delivering iterative, reviewable progress throughout the academic year.

The advanced artefact submitted with this report demonstrates the core functionality of the platform across all three user roles which Consumer, Service Provider, and Administrator which and has achieved a System Usability Scale score of 78.5 in initial user acceptance testing. Identified deviations from the original proposal, most notably the partial implementation of digital payment integration, have been managed transparently and remain on the development backlog for completion prior to the final submission.

HomeFix demonstrates that responsible, professional software engineering involves far more than technical competence: it requires deliberate consideration of the social impacts of the system being built, rigorous ethical commitments to user data sovereignty and fairness, compliance with applicable legal frameworks, and a security architecture that protects users from a wide range of technical threats. Each of these dimensions has been addressed in both the design philosophy and the concrete implementation of the HomeFix platform.

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Appendices

Appendix A: Sprint Log Summary

Sprint	Period	Key Deliverables	Status
Sprint 1	Oct – Nov 2025	Requirements analysis, literature review, system design, Figma prototypes	Complete
Sprint 2	Nov – Dec 2025	Database schema design, backend scaffolding, authentication module	Complete
Sprint 3	Jan 2026	Frontend homepage, search, provider profile components	Complete
Sprint 4	Feb 2026	Booking lifecycle management, review system, admin dashboard	Complete
Sprint 5	Mar 2026	Testing, debugging, advanced artefact, draft report	In Progress
Sprint 6	Apr – May 2026	Digital payment integration, final testing, final report, poster, viva	Planned

Appendix B: API Endpoint Reference

Method	Endpoint	Description	Auth Required
POST	/api/auth/register	Register new user (consumer or provider)	No
POST	/api/auth/login	Authenticate user, issue JWT	No
GET	/api/providers/search	Search/filter providers by keyword and criteria	No
GET	/api/providers/:id	Get full provider profile	No
POST	/api/bookings	Create a new booking request	Consumer
PUT	/api/bookings/:id/status	Update booking status	Provider/Admin
GET	/api/bookings/my	Get current user's booking history	Consumer/Provider
POST	/api/reviews	Submit a review for a completed booking	Consumer
GET	/api/admin/providers	List all providers (admin)	Admin
DELETE	/api/admin/users/:id	Remove a user account (admin)	Admin

Appendix C: Necessary Links

GitHub Repository: https://github.com/prakashbhatt7/HomeFix.git

Figma Prototype: https://www.figma.com/proto/p8ywRdr9HYSNfEwZRFunnv/Web-App?node-id=0-1&t=qk6Lwqd8sPYw6cDm-1